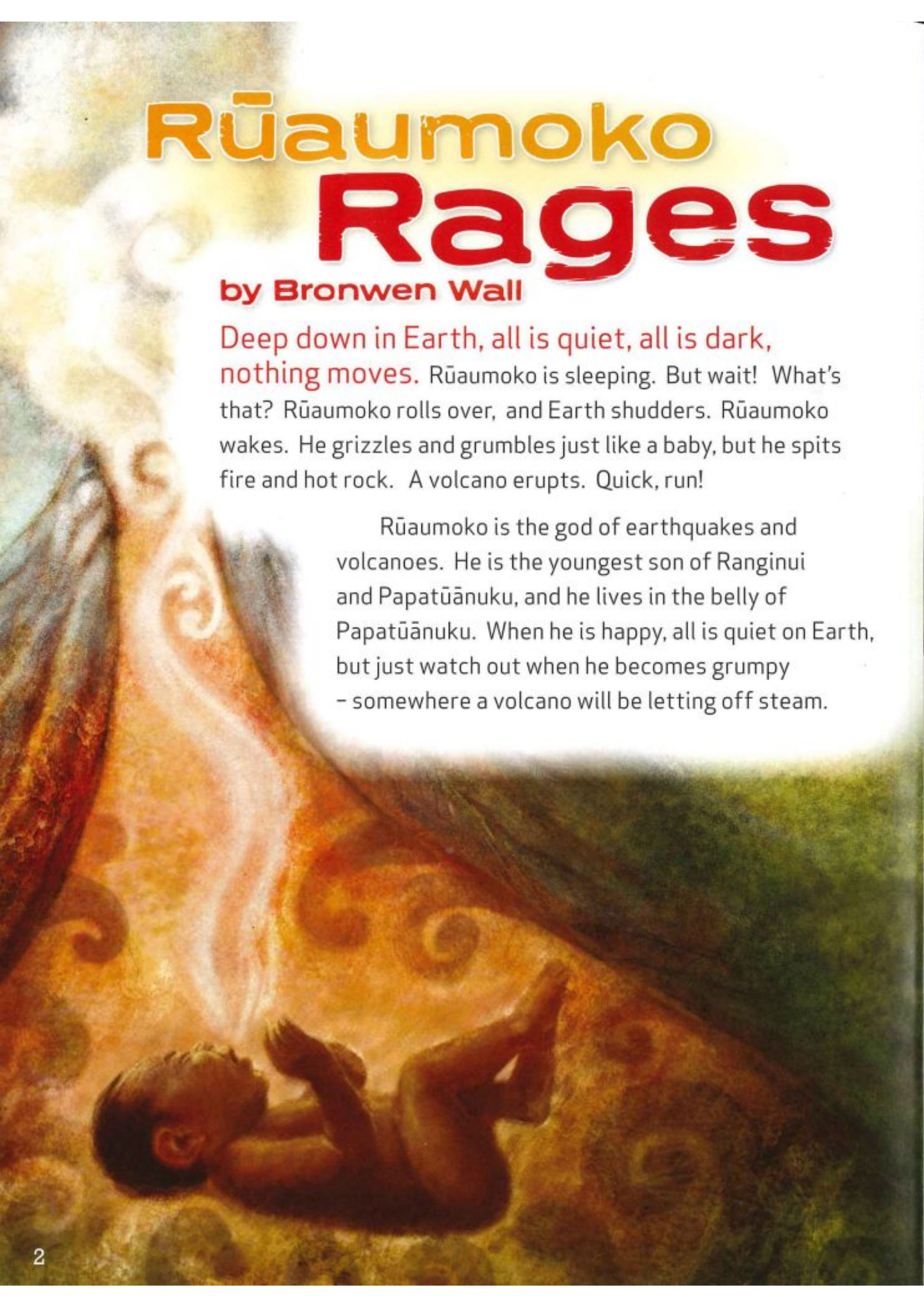


The background of the page is a painting. At the bottom, a baby with dark skin is lying on its back on a surface with a swirling, shell-like pattern in shades of orange and brown. The baby's arms are raised, and its legs are bent. In the upper left, a volcano is erupting, with a large plume of white smoke or steam rising into the sky. The overall scene is set against a bright, hazy sky.

# Rūaumoko Rages

by Bronwen Wall

Deep down in Earth, all is quiet, all is dark, **nothing moves**. Rūaumoko is sleeping. But wait! What's that? Rūaumoko rolls over, and Earth shudders. Rūaumoko wakes. He grizzles and grumbles just like a baby, but he spits fire and hot rock. A volcano erupts. Quick, run!

Rūaumoko is the god of earthquakes and volcanoes. He is the youngest son of Ranginui and Papatūānuku, and he lives in the belly of Papatūānuku. When he is happy, all is quiet on Earth, but just watch out when he becomes grumpy – somewhere a volcano will be letting off steam.



A volcano is a vent, or hole, that leads far into Earth. It runs down to the magma found deep in the Earth's crust in the **mantle**. Magma is a mixture of **molten** rocks and gases that has not reached the Earth's surface. These rocks and gases are very hot – about four times hotter than a pizza oven. Sometimes, magma is pushed up into spaces in Earth's crust called magma chambers. There, the magma bubbles and plops inside these chambers like thick soup in a pot.

## Beneath our Feet

### Crust

Earth is covered by a lumpy, bumpy crust. It is made up of large slabs of solid rock. Some of the crust is the land we live on. The crust is under the ocean as well.

### Mantle

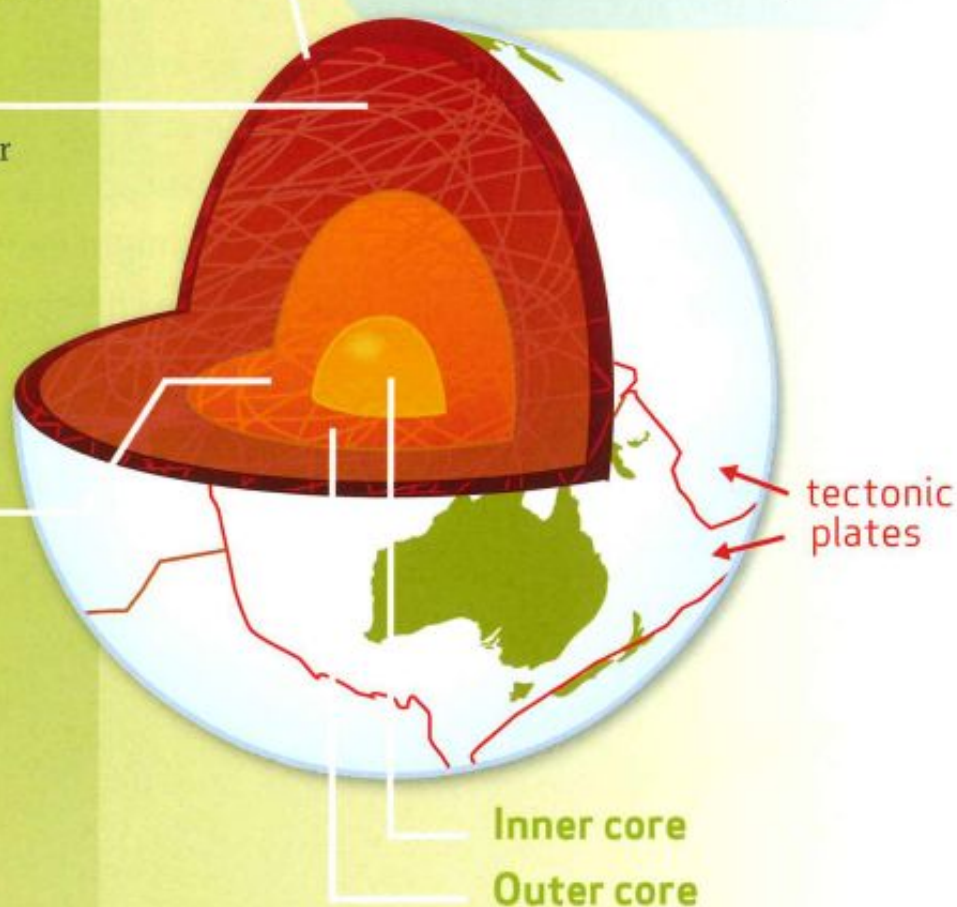
Below the crust is a much thicker layer called the mantle. The deeper you go into the mantle, the hotter things get. The mantle becomes more liquid as it gets hotter, but it's not as hot as Earth's core.

### Core

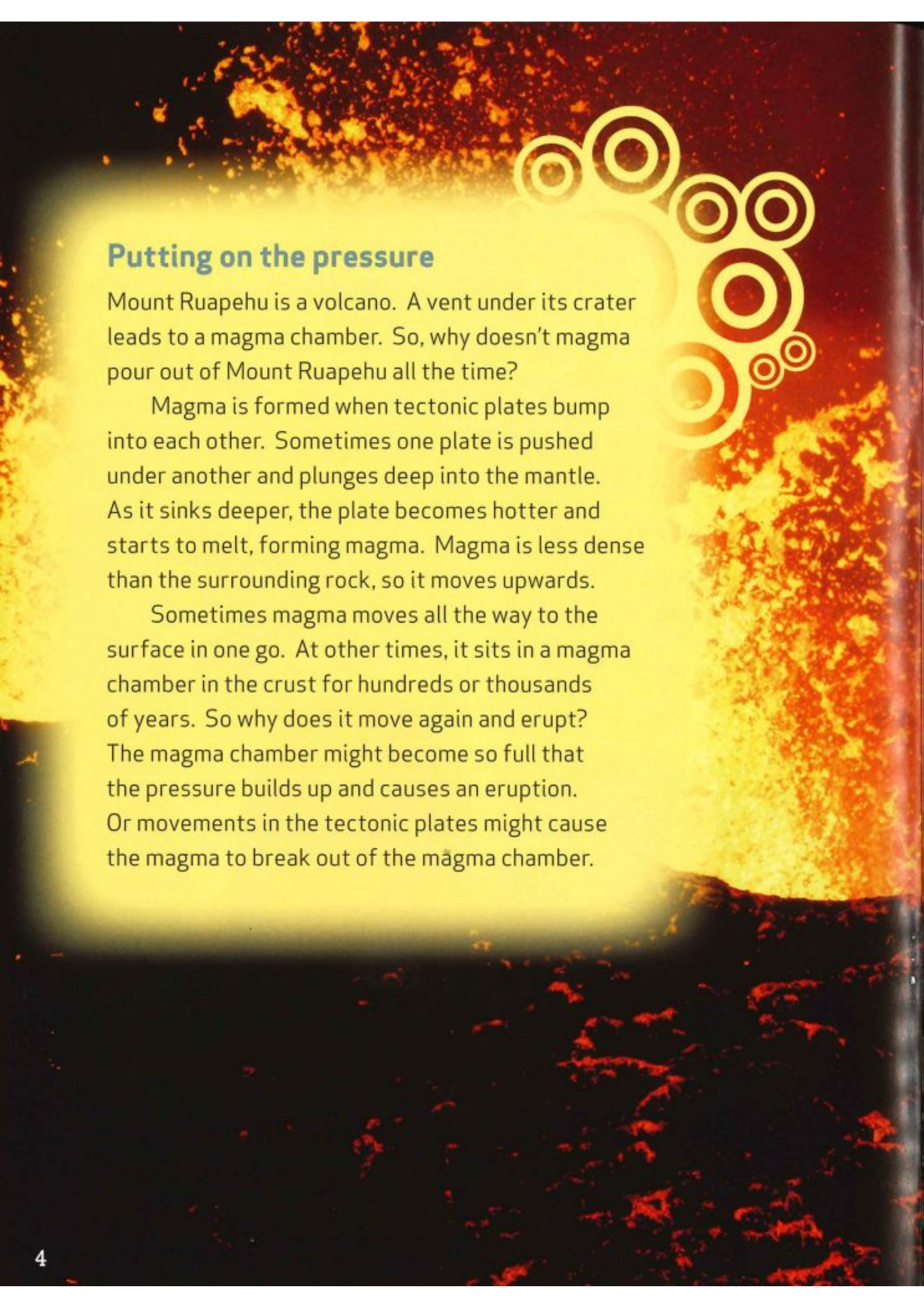
Earth's core is made up of an outer core, which is liquid, and an inner core, which is solid.

### Tectonic plates

Large slabs of the crust and the top part of the mantle make up tectonic plates. The plates float above the lower part of the mantle and move slowly around.







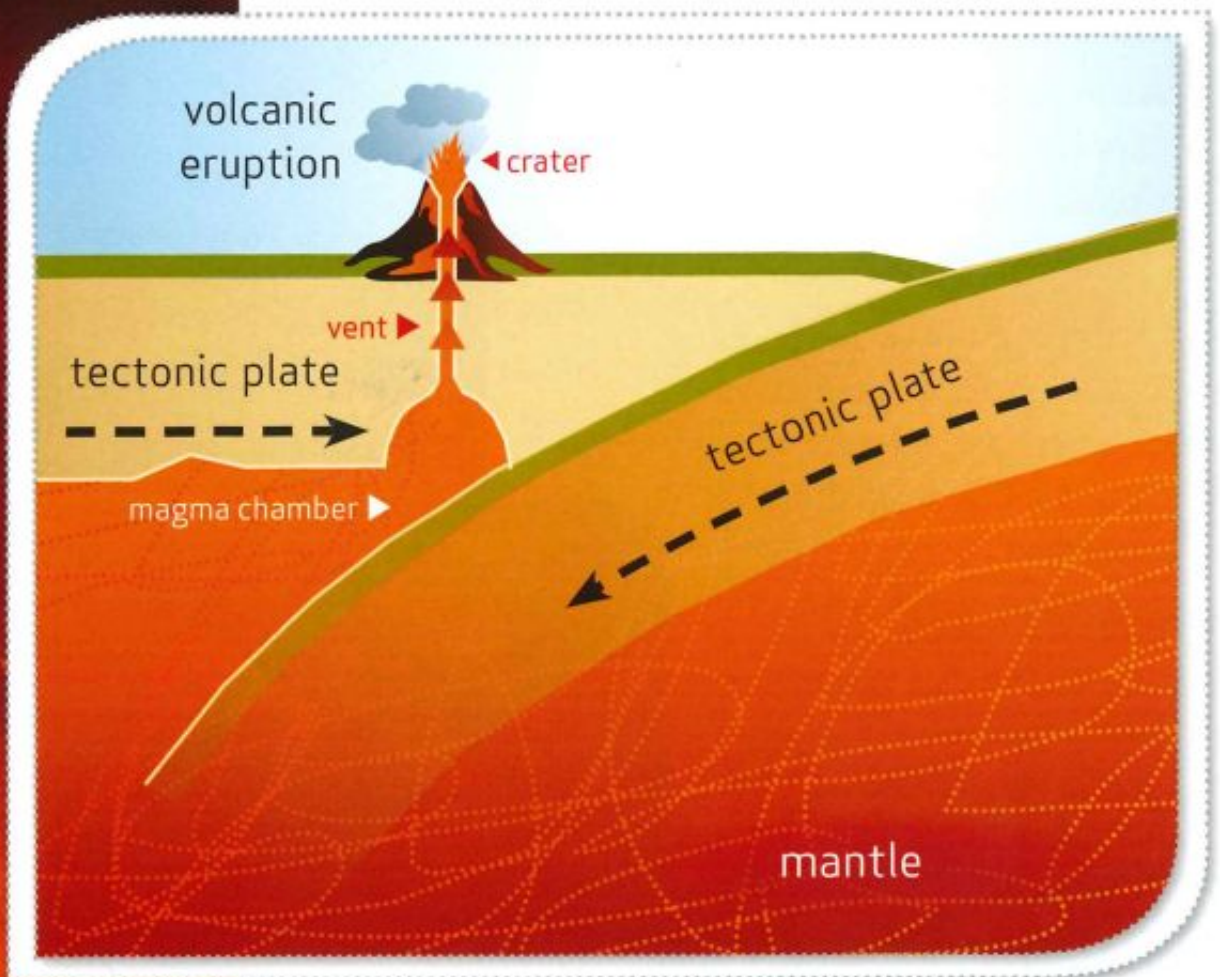
## Putting on the pressure

Mount Ruapehu is a volcano. A vent under its crater leads to a magma chamber. So, why doesn't magma pour out of Mount Ruapehu all the time?

Magma is formed when tectonic plates bump into each other. Sometimes one plate is pushed under another and plunges deep into the mantle. As it sinks deeper, the plate becomes hotter and starts to melt, forming magma. Magma is less dense than the surrounding rock, so it moves upwards.

Sometimes magma moves all the way to the surface in one go. At other times, it sits in a magma chamber in the crust for hundreds or thousands of years. So why does it move again and erupt? The magma chamber might become so full that the pressure builds up and causes an eruption. Or movements in the tectonic plates might cause the magma to break out of the magma chamber.





Once the magma erupts from a volcano, it is called lava. Lava can be thick and slow-moving or runny and fast-moving. It will be anything from 700 to 1200 degrees Celsius. (Water boils at 100 degrees Celsius.) It will be glowing red or even white with the heat. Over time, as the lava cools, it becomes solid rock.



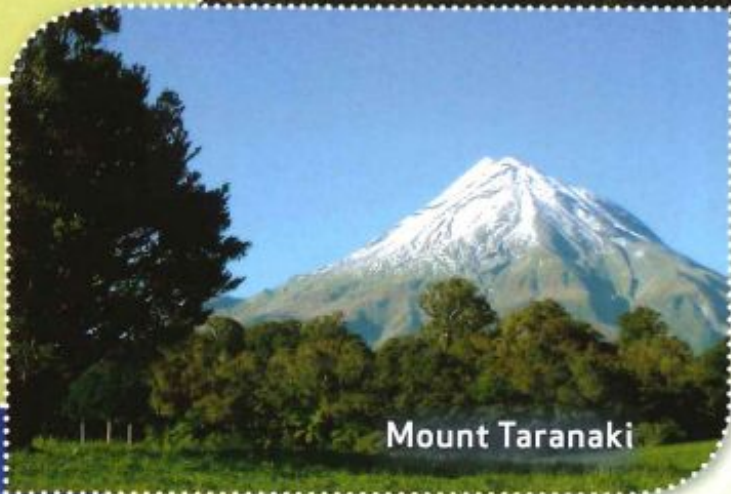
# Active, dormant, extinct

Volcanoes can be active, dormant, or extinct.

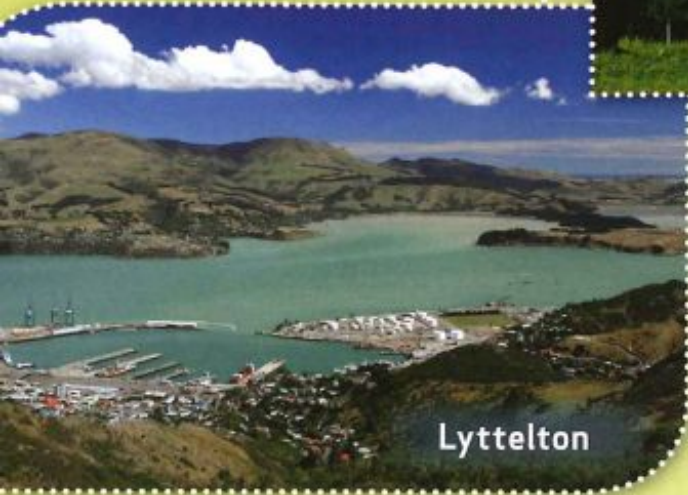
- **Active** volcanoes erupt often. They may have erupted fairly recently, and we can expect them to erupt at any time.
- **Dormant** volcanoes appear to be sleeping. It's been a while since they last erupted, but we think that they could still erupt at any time.
- **Extinct** volcanoes haven't erupted for such a long time that they probably won't ever again.



Mount Ruapehu



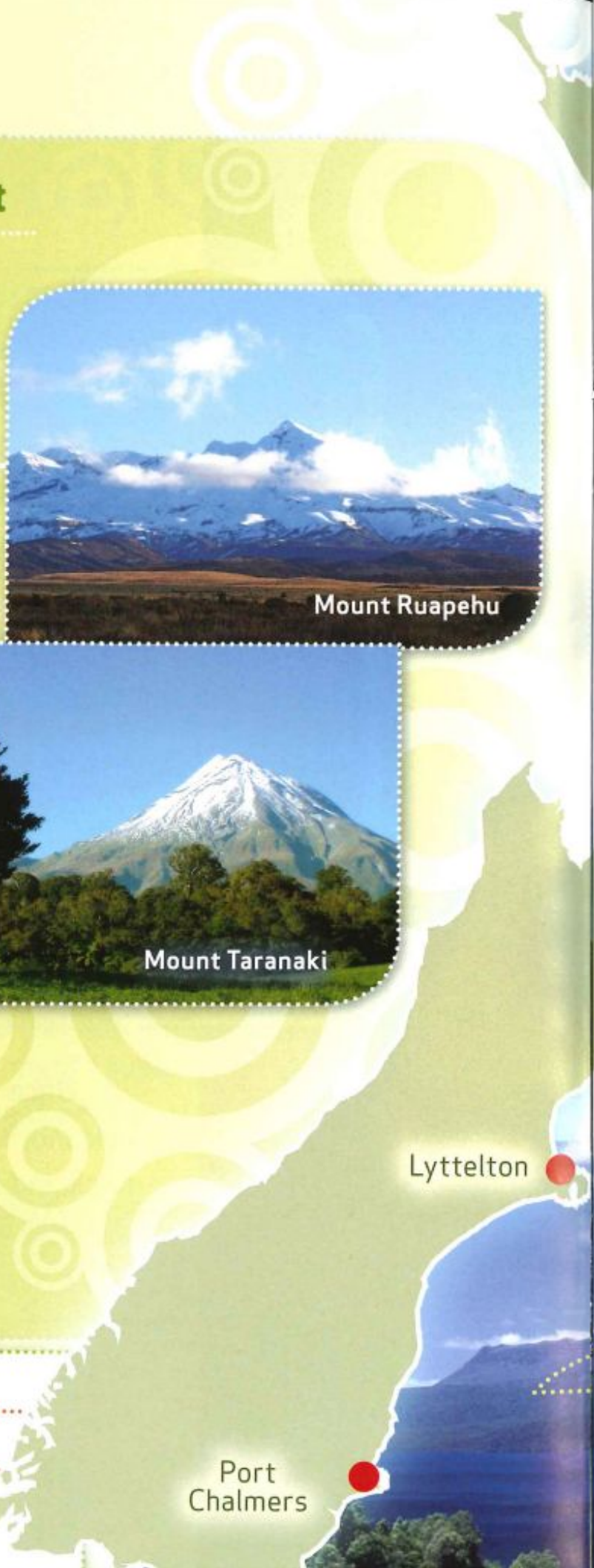
Mount Taranaki



Lyttelton

## Key

- Lava dome volcanoes
- ▲ Strato cone volcanoes
- Shield volcanoes



Port Chalmers



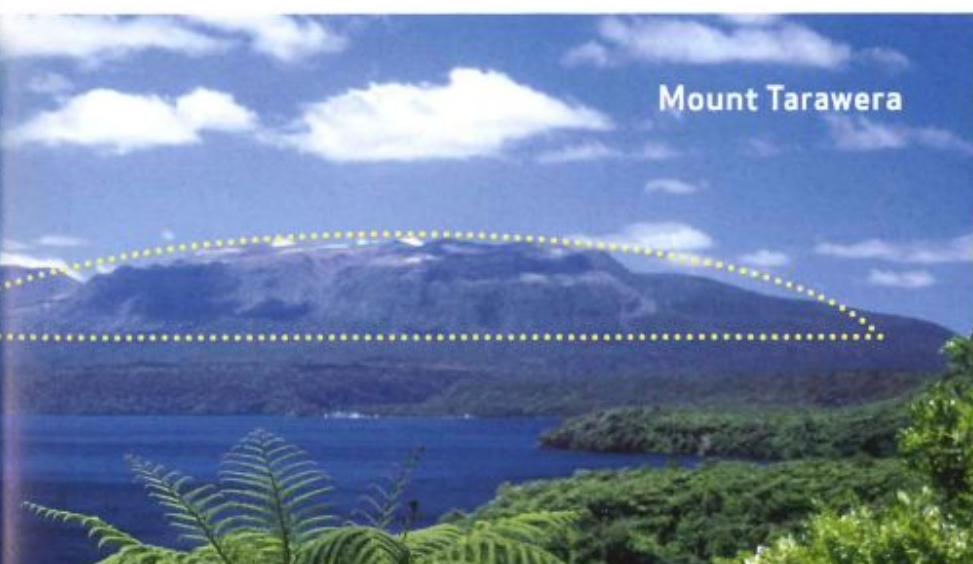


## It takes all kinds

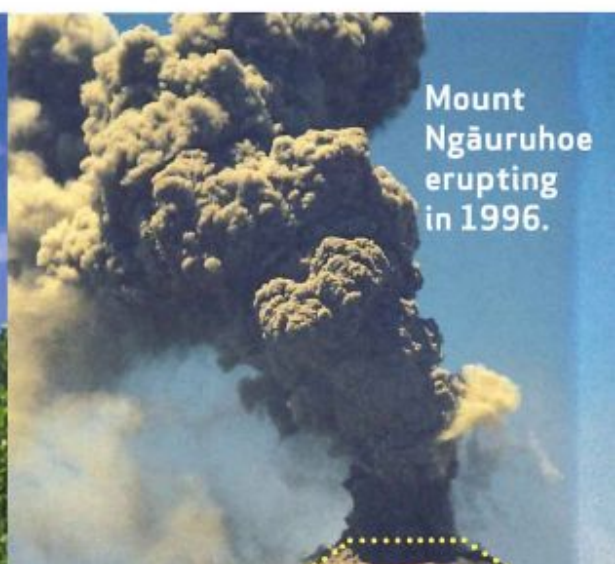
Volcanoes come in all shapes and sizes – lava domes, strato cones, shields, calderas, or fields. They can be active, dormant, or extinct. In New Zealand, there are a range of different types.

A lava dome volcano occurs when the lava flows slowly from a volcano's vent and piles up around the vent in a rounded bulge. Mount Tarawera and Mount Ngongotahā near Rotorua are lava dome volcanoes.

Mount Taranaki, Mount Ngāuruhoe, and Mount Ruapehu are strato cone volcanoes. They look like upside-down ice cream cones. These volcanoes form when the eruption is violent and lava is flung high into the sky. As it shoots into the cold air, the lava cools quickly and becomes **cinder**. The cinder breaks up and falls like rain back to the ground around the volcano's vent. Lava also flows down the sides, building a steep pile that becomes a cone shape.



Mount Tarawera



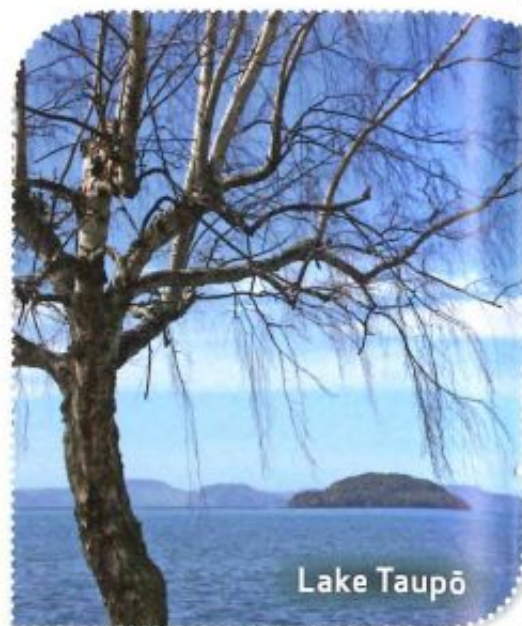
Mount Ngāuruhoe erupting in 1996.



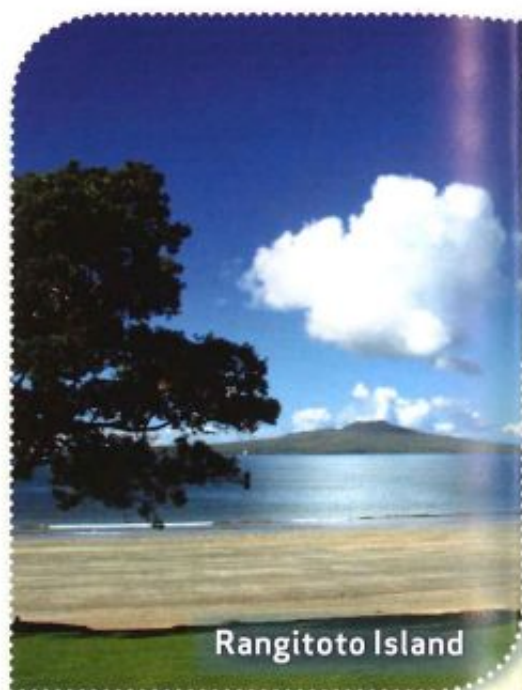
Lake Taupō and Lake Rotorua are caldera volcanoes. "Caldera" is a Spanish word. It means cauldron, which is a big pot. A caldera forms when an eruption is so big that the ground collapses once the magma chamber is emptied. Later the hole often fills with water to become a lake.

Auckland city sits right on top of a volcanic field. That's a spot where Earth's crust is weak or has cracks. Magma wells up to the surface from the mantle without sitting in a chamber first. Hot spots normally form into fields of volcanoes. There are about 49 small volcanoes in the field that covers Auckland. The last volcano in this field erupted at Rangitoto 600 years ago. That's not a long time in terms of a volcano, so we think that this field is still active.

The only type of active volcano we don't have in New Zealand is a shield volcano, although we have some large, extinct ones. Shield volcanoes don't often have explosive eruptions. Instead the lava flows slowly down the sides of the vent, forming a gentle slope. Lyttelton in Christchurch and Port Chalmers in Dunedin are built on extinct shield volcanoes.



Lake Taupō



Rangitoto Island

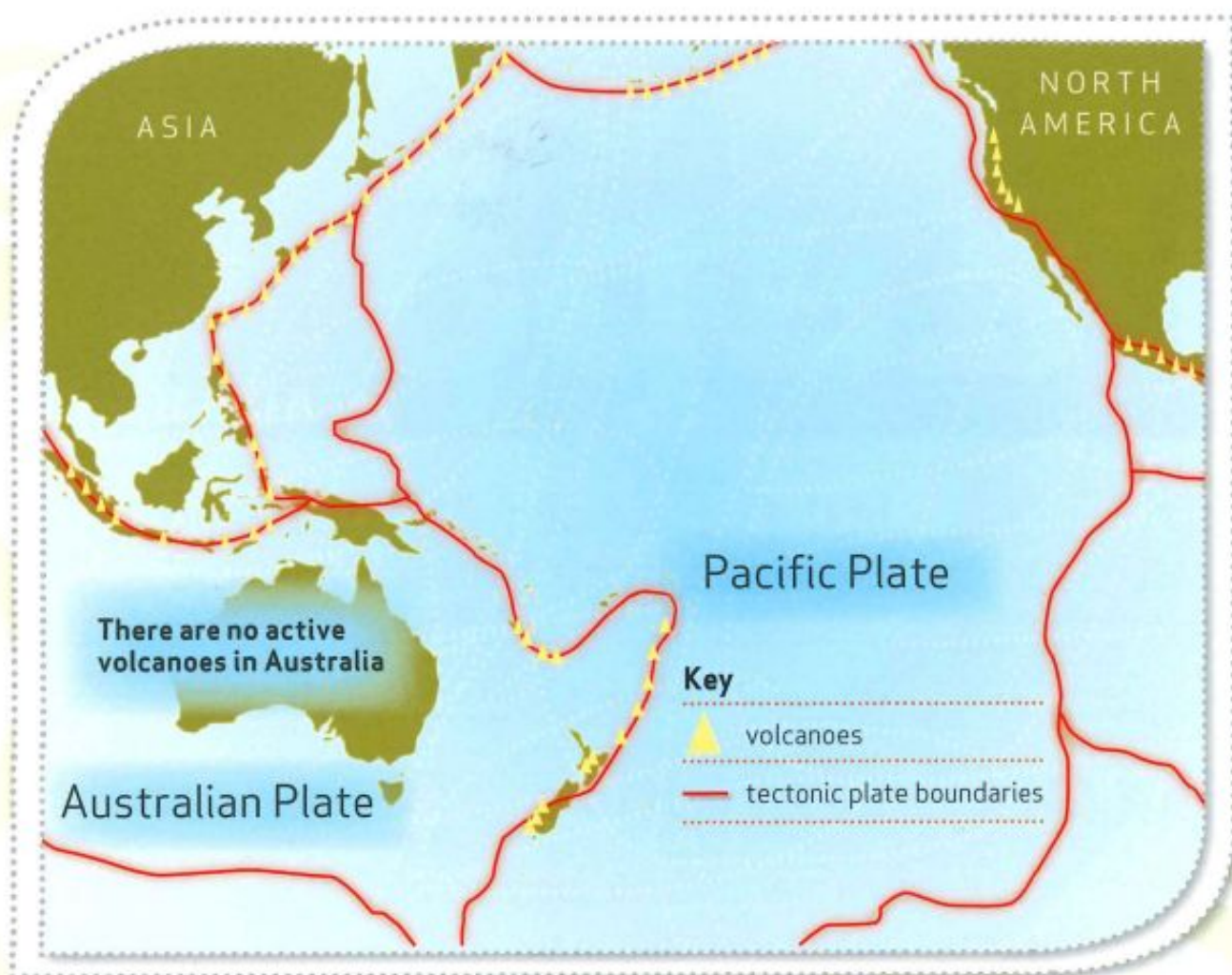


Port Chalmers



## The Ring of Fire

Most of the world's active volcanoes are found along the edges of the tectonic plates. More than half of them lie around the edges of the Pacific Plate. That's why this area is called the Ring of Fire. And New Zealand sits right on the edge of the Australian and the Pacific plates in the Ring of Fire. That makes for a lot of hot rock in one small country.



## Glossary

- cinder** – materials that have been thrown from a volcano
- mantle** – a layer of hot melted rock
- molten** – materials that have been melted by the heat